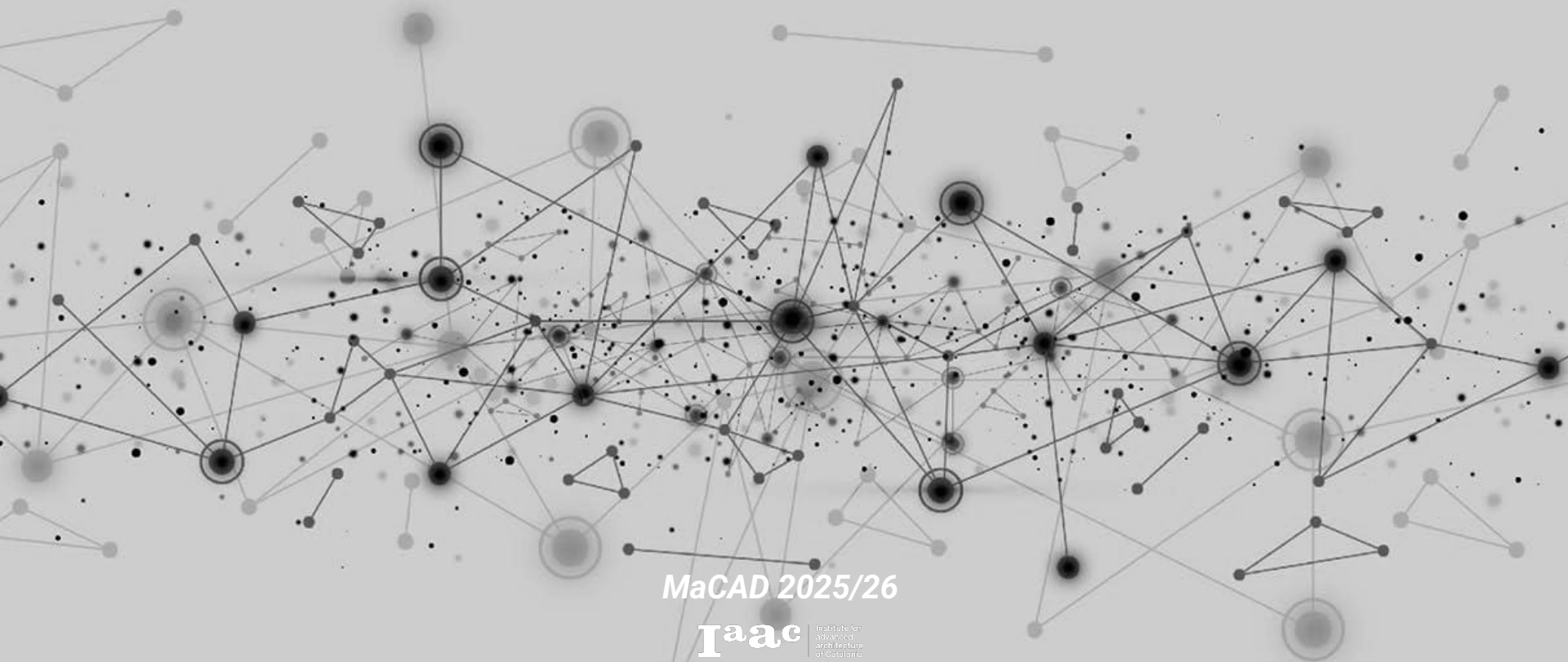


# Graph Machine Learning Seminar

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Graph Machine Learning: Learn about the latest advancements in graph data to build robust machine learning models



MaCAD 2025/26

Iaac Institute for  
Advanced  
Architecture  
of Catalonia

## Assignment 03 : Graph Machine Learning for Architectural Data

Creating BGR Graph



### **What This Assignment Is About?**

How to understand Machine Learning through Architecture.

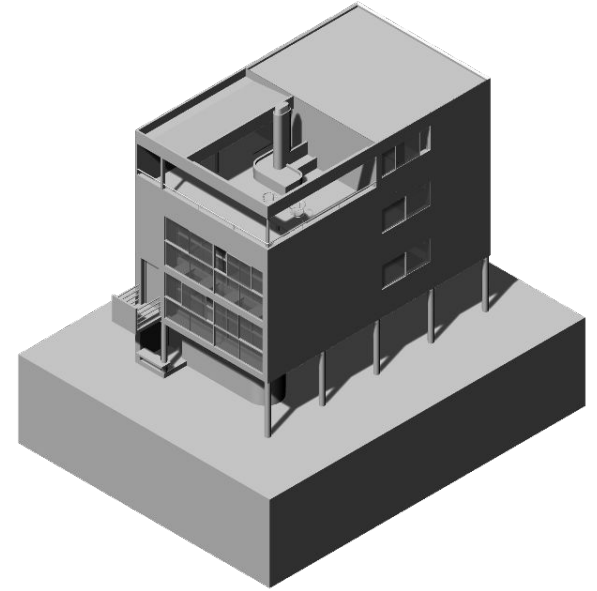
The Data set for this assignment is called BGR (Building-Ground Relationship) and from the Notebook it classifies how a building sits on the ground into 5 types:

- 0 Separation (building floats above ground)
- 1 Separation with Plinth
- 2 Adherence (building sits directly on ground)
- 3 Adherence with Plinth
- 4 Interlock (building is partially embedded in ground)

## Model Choice

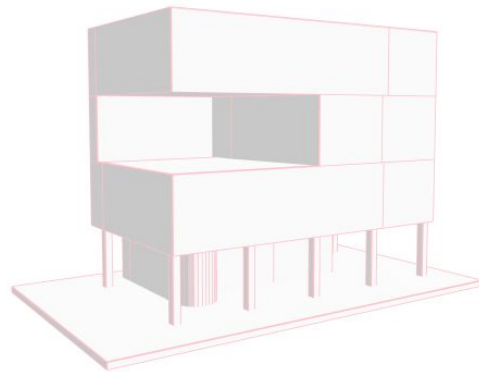
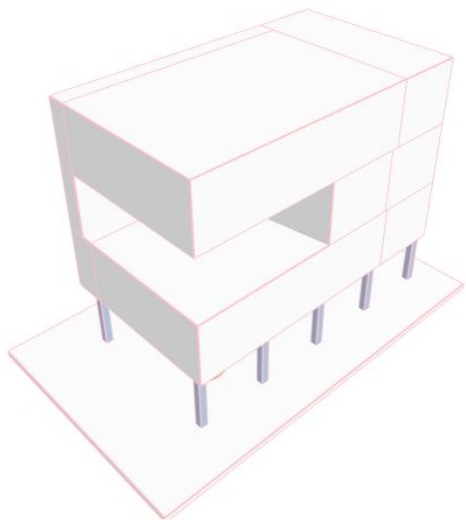
For this Assignment we find adequate to analyse one of the buildings of **Le Corbusier**, for its interesting relationship with the ground and in particular **Maison Citrohan (1922)**.

The building sits on pilotis, so it is lifted off the ground, which is separation with Plinth



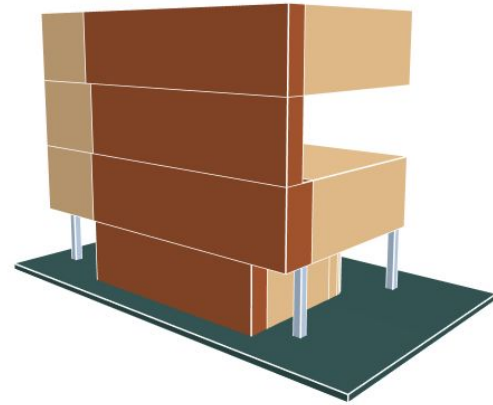
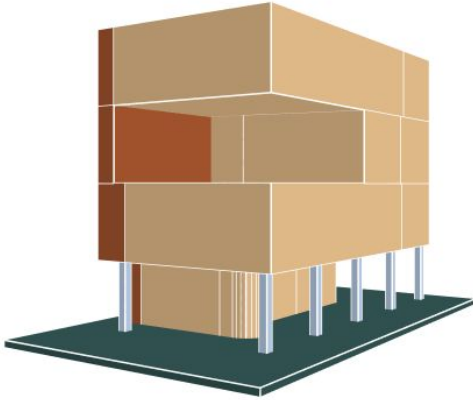
## Creating BGR Graph

Model has a double Height that we simplified and we left the curved entrance in the ground floor just to maintain the Le Corbusier Style



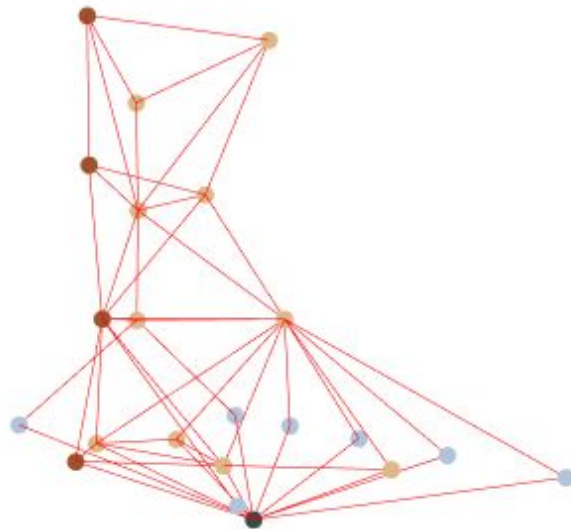
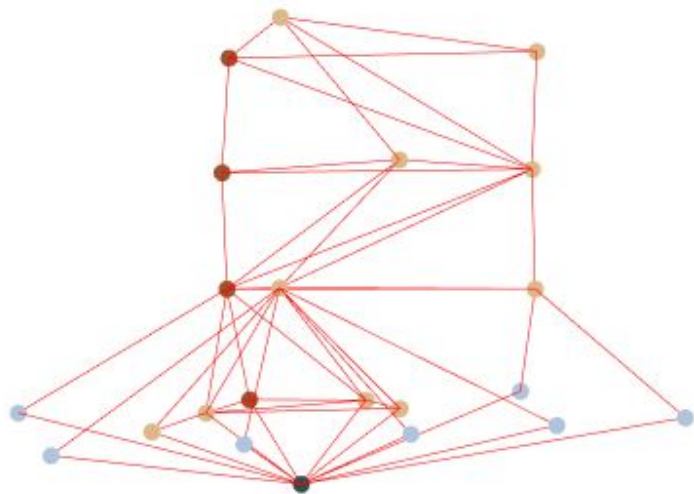
## Creating BGR Graph

We assigned Colors to each Cell



## Creating BGR Graph

We generated the adjacency Graph



## Prediction

After training the model, the prediction is on-point

### 6. Predict the dataset

Generate Code Markdown

```
pred_results = pyg_2.Predict()
predicted_values = pred_results['pred'].tolist()
predicted_labels = [MAPPING[pv] for pv in predicted_values]
actual_values = pred_results['y_true'].tolist()
actual_labels = [MAPPING[av] for av in actual_values]
probabilities = pred_results['prob'].tolist()
```

```
df = pd.DataFrame({
    "Actual Value": actual_values,
    "Predicted Value": predicted_values,
    "Actual Label": actual_labels,
    "Predicted Label": predicted_labels,
    "Confidence": [round(max(p), 2) for p in probabilities]
})
```

df

[9] ✓ 0.1s

Python

| Actual Value | Predicted Value | Actual Label | Predicted Label        | Confidence |
|--------------|-----------------|--------------|------------------------|------------|
| 0            | 0               | 1 Separation | Separation with Plinth | 1.0        |